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Paleomagnetic dating of Alleghanian orogenesis and mineralisation in the Mascot–Jefferson City zinc district of East Tennessee, USA

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Abstract

The Mascot–Jefferson City (M-JC) Mississippi Valley-type (MVT) deposits are in the Valley and Ridge province of the Appalachian orogen in East Tennessee. They have been a major source of zinc for the USA but their age is uncertain and thus their genesis controversial. About 10 specimens from each of 37 sites have been analysed paleomagnetically using alternating field and thermal step demagnetisation methods and saturation isothermal remanence methods. The sites sample limestones, dolostones, breccia clasts and sphalerite–dolomite MVT mineralisation from mines in the Lower Ordovician Kingsport and Mascot formations of the Knox Group. The characteristic remanent magnetisation (ChRM) is carried by magnetite in the limestones, by both magnetite and pyrrhotite in the dolostones and by pyrrhotite preferentially to magnetite in the mineralisation. Mineralized sites have a more intense ChRM than non-mineralised, indicating that the mineralising and magnetisation event are coeval. Paleomagnetic breccia tests on clasts at the three sites are negative, indicating that their ChRM is post-depositional remagnetisation, and a paleomagnetic fold test is negative, indicating that the ChRM is a remagnetisation, and a post-dates peak Alleghanian deformation. The unit mean ChRM direction for the: (a) limestones gives a paleopole at 129°E, 12°N ($d_p = 18°$, $d_m = 26°$, $N = 3$), indicating diagenesis formed a secondary chemical remanent magnetisation during the Late Ordovician–Early Silurian; (b) dolomitic limestones and dolostone host rocks gives a paleopole at 125.3°E, 31.9°N ($d_p = 5.3°$, $d_m = 9.4°$, $N = 7$), recording regional dolomitisation at 334 ± 14 Ma (1σ); and (c) MVT mineralisation gives a paleopole at 128.7°E, 34.0°N ($d_p = 2.4°$, $d_m = 4.4°$, $N = 25$), showing that it acquired its primary chemical remanence at 316 ± 8 Ma (1σ). The mineralisation is interpreted to have formed from hydrothermal fluid flow, either gravity or tectonically driven, after peak Alleghanian deformation in eastern Tennessee with regional dolomitisation of the host rocks occurring as part of a continuum during the ~ 20 Ma prior to and during peak deformation. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Paleomagnetism; Alleghanian orogeny; Mascot–Jefferson City ore; Age dating

1. Introduction

The Mascot–Jefferson City Mississippi Valley-type (MVT) zinc deposits are located in the Valley and Ridge physiographic province of the Appalachian orogen in eastern Tennessee (Fig. 1). They have been in fairly continuous production since 1854, and were the main source of zinc for the USA from the mid-1950s to about 1990, providing more than half of its production at times (Fulweiler and Hill, 1992).

The Valley and Ridge Province is bounded on the northwest by the Appalachian Plateau and on the

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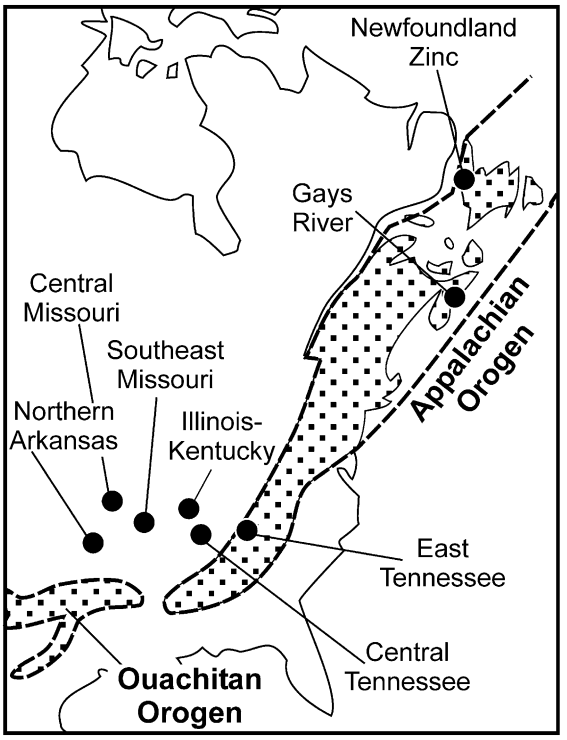


Fig. 1. Location of East Tennessee and other Mississippi Valley-type Pb–Zn–Ba–F districts in the Appalachian orogen and adjacent midcontinental USA that have been studied paleomagnetically (see Symons et al., 1996).

southeast by the Blue Ridge (Fig. 2). The province comprises an ~3.5-km thick sequence of Lower Cambrian to Mississippian carbonate and clastic sediments that have been disrupted by a series of late Paleozoic Alleghanian NE–SW-trending thrust faults (Hatcher and Odom, 1980; Misra and Fulweiler, 1992). These faults record thrusting from the SE to NW with tens of kilometers of displacement, mostly in the Paleozoic strata rather than the underlying Precambrian basement in the Tennessee Valley area (Rodgers, 1949).

MVT Zn ± Pb mineralisation of economic interest in eastern Tennessee occurs in strata ranging from the Lower Cambrian and it occurs in several definable districts of which the Mascot–Jefferson City (M-JC) district is the most important (Fig. 2) (Misra et al., 1983). Based on a palinspastic reconstruction, Roeder and Witherspoon (1978) suggested that the M-JC and Copper Ridge districts, now separated by the Saltville thrust fault, formed one large district prior to the Alleghanian orogeny. Also, the Sweetwater MVT district to the southwest but along strike with the M-JC district is in a very similar geologic setting, but the mineralisation is dominantly fluorite and barite with minor sphalerite (Misra et al., 1989).

A major controversy swirls around the age of the M-JC MVT deposits. Geologic and direct and indirect radiometric dating methods have given answers ranging from Early Ordovician to mid-Permian with little agreement between the methods. Two other controversies are closely related. One is the age of dolomitisation of the host rocks and the second is the genesis of the mineralisation. This paper reports paleomagnetic data for a large collection of specimens from MVT mineralisation and host rocks, taken from four mines spanning the M-JC district, which provides direct dates for both the mineralisation and dolomitisation events.

2. Geology

The following discussion of the M-JC district is intended to provide only a brief overview of most aspects of the geology; its emphasis is on factors that relate to the dating of mineralisation and dolomitisation.

2.1. Stratigraphy

The Paleozoic section in the M-JC district is about 3.5 km thick and ranges in age from Early Cambrian to early Mississippian (Hill, 1971; Walker and Lutz, 1992). Most of the economic MVT mineralisation has come from several units near the contact between the Lower Ordovician Kingsport and Mascot formation of the Knox Group (Table 1). Most of the Kingsport Formation comprises dark to light-gray fine-grained dolostones, but its upper third, which hosts most of the ore, is comprised of dove gray to brown limestone with medium to coarse-grained dolostone interbeds (Harris, 1969). Above, in the Mascot Formation, are light and dark-gray fine-grained dolostones that are moderately cherty with several ore units in the bottom fifth of the formation. In general, the main ore beds are somewhat lower stratigraphically in the northeast-

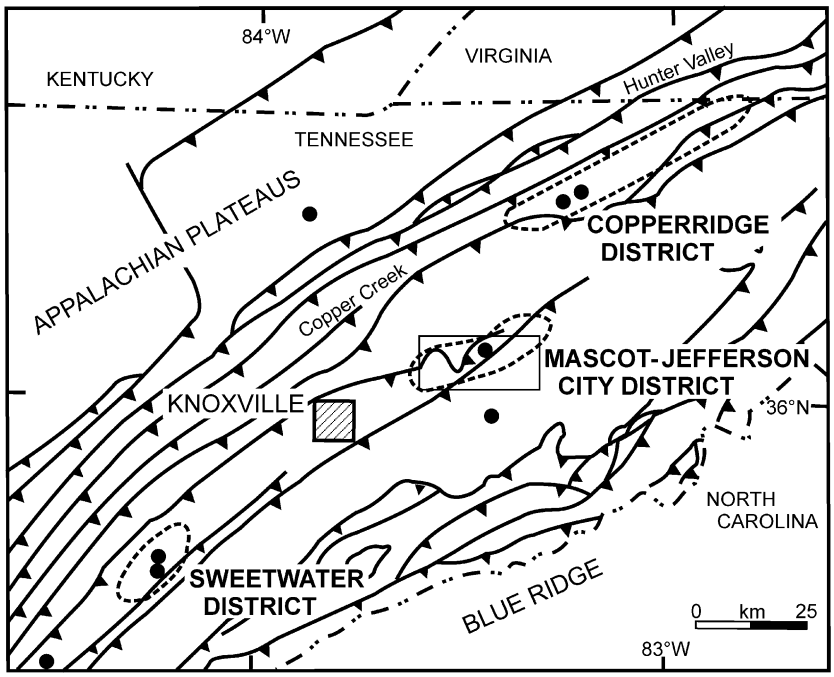


Fig. 2. Location of the Mascot–Jefferson City district in the Valley and Ridge Province of the Appalachian orogen.

ern part of the M-JC district than in the southwestern part (Fig. 3; Hoagland et al., 1965).

2.2. Dolomitisation

The fine-grained dolostones are regionally persistent and contain xenotropic crystals of <0.06 mm in size. They are thought to be essentially syngenetic and primary in origin, having formed in a subtidal, intertidal or supratidal environment (Laporte, 1971; Harris, 1971, 1973; Churnet and Misra, 1981; Churnet et al., 1982; Haynes and Kesler, 1994). If true, then these fine-grained dolostones are Early Ordovician in age. The medium to coarse-grained dolostones contain dolomitic crystals of >0.06 mm in size. The crystals are commonly porphyrotopic and typically zoned (Misra and Fulweiler, 1992). These coarser grained dolostones have been interpreted to be diagenetic (Kendall, 1960; Churnet and Misra, 1981), to be epigenetic but pre-mineralisation (Hoagland, 1976; Montanez, 1994; Haynes and Kesler, 1994), to predate some but not all mineralisation because sphalerite often replaces dolomite rhombs in dolomitised limestones (Matlock and Misra, 1993), to be syn-mineralisation (Haynes et al., 1989), and to be both syngenetic and syn-mineralisation (Oder and Ricketts, 1961; Hoagland, 1976). Thus, depending on the interpretation, any age from Middle Ordovician to Permian is possible for the genesis of the medium to coarse-grained dolostones.

2.3. Structure

The M-JC MVT district covers an area of about 50×25 km (Fig. 3). It is subdivided into three segments by the Rocky Valley and Mill Springs faults. These are parallel, NE–SW trending, northwest-verging thrust faults that are deemed to be Alleghanian in age (Misra et al., 1983). The intervening thrust sheets have been deformed into gently dipping synclinal/basinal and anticlinal/domal structures by minor cross-structures. They have amplitudes and wavelengths of low hundreds and low thousands of meters, respectively. The cross-structures do not appear to have played a role in localizing the zinc mineralisation, implying that it predates the Permian end of the Alleghanian orogeny (Hoagland et al., 1965). They result, however, in the Immel Mine strata dipping

Table 1
Generalized stratigraphic column for the Mascot–Jefferson City zinc district (after Hill, 1971; Walker and Lutz, 1992)

Age	Unit and lithology	Thickness (m)
Mississippian to Ordovician	post-Chickamauga sedimentary rocks	~ 1100
Middle	Chickamauga Group—limestone	420–760
	unconformity	
Lower	Knox Group	
	Mascot Formation—cherty dolostone and limestone	75–130
	Kingsport Formation—dolostone and limestone	50–150
Cambrian		
Upper	Chepultepec Formation—dark-gray dolostone and limestone	180–230
	Copper Ridge Formation—cherty dolostone	320–370
Middle	Conasauga Group—shale, limestone and dolostone	~ 650
Lower	Rome Formation—shale	~ 370
	Shady Formation—dolostone	~ 300
	unconformity	
Precambrian	basement	

north-northwesterly away from the Strawberry Plains anticline, the Young Mine strata dipping in all directions away from the Hodges dome, the New Market mine strata straddling the axis of the New Market syncline, and the Coy Mine strata dipping southwesterly away from the Cherokee anticline, with the site attitudes shown in Table 2.

2.4. Brecciation

The M-JC ore-bearing structures consist of two interrelated types of breccia bodies. "Breakthrough" structures cut one or more stratigraphic units. They are discordant bodies up to several thousand meters long, up to a few hundred meters wide, and up to >150 m high (Hoagland et al., 1965; Crawford et al., 1969). They may be V-shaped, low-domal or high-domal structures that appear to follow pre-mineralisation faults and joints, and that are filled with a rubble breccia containing several petrographic types of clasts (McCormick et al., 1971; Hill, 1969; Hill et al., 1971a; Hoagland, 1976). Crackle breccias are found around the top and sides of the breakthrough structures. The Immel, Young and New Market ore bodies are in "breakthrough" breccias. "Bedded" structures occur only within certain limestone beds in the Kingsport Formation. They are semi-concordant tabular mantos of dolomitisation, brecciation and mineralisation that appear to extend laterally from breakthrough breccia bodies in most cases (Kendall, 1960; Fulweiler and McDougal, 1971). These bedded bodies rarely exceed more than tens of meters wide in length

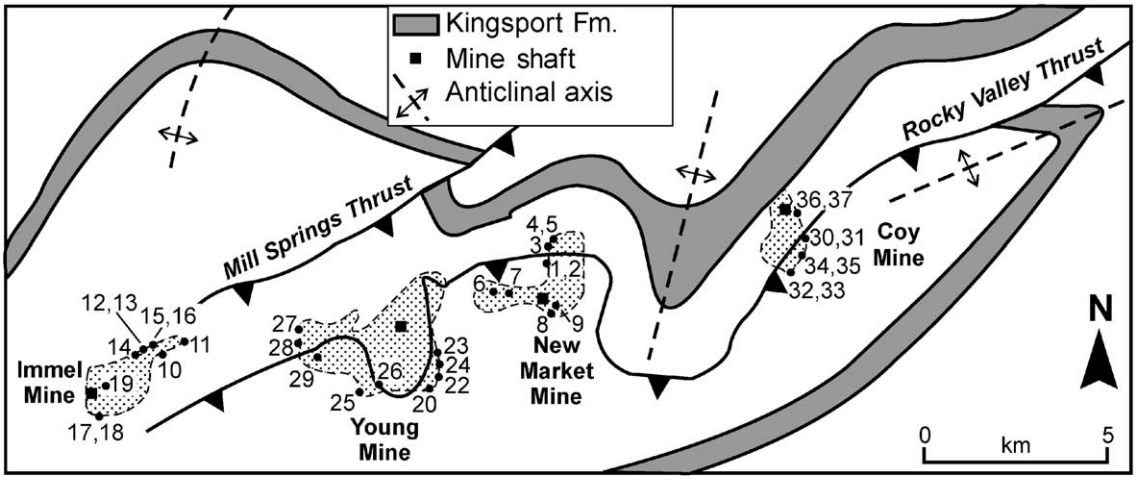


Fig. 3. Simplified geology of the Mascot–Jefferson City district with the locations of the mines and sampling sites used in this study (adapted from Hoagland, 1976; Misra and Fulweiler, 1992, and mine maps of ASARCO ca. 1997).

Table 2
Summary of site mean remanence data

Site	Sample	Str, Dip	Demagnetisation		N, N_d, N_c	In situ ChRM direction			
			AF (mT)	TM (°C)		Dec (°)	Inc (°)	k	α_{95} (°)
1	O,N	082, 08	20–80	80–600	14, 10, 0	–	–	–	–
2	dL,N	185, 09	20–80	80–500	17, 10, 8	146	22	90	6
3	O,N	082, 05	20–80	80–500	19, 10, 8	154	26	95	6
4	O,N,bxt	– , –	20–100	80–450	31, 17, 17	152	31	53	5
5	L,N	260, 14	20–110	80–450	21, 10, 8	146	39	264	3
6	O,N	010, 04	20–120	80–535	22, 10, 9	152	31	89	5
7	O,N	070, 07	20–105	80–500	20, 10, 10	149	31	40	8
8	O,N	250, 03*	20–120	80–500	16, 10, 10	154	24	63	6
9	O,N	250, 03*	20–120	80–500	17, 10, 10	151	31	134	4
10	O,I	282, 28	20–120	80–450	22, 11, 10	162	13	87	5
11	L,I	285, 23	20–135	80–500	13, 10, 0	–	–	–	–
12	O,I	285, 28	10–115	80–535	18, 10, 8	160	20	47	8
13	dL,I	256, 30	20–120	80–500	18, 10, 10	160	18	115	5
14	O,I	273, 26	20–120	80–535	16, 10, 10	163	14	89	6
15	O,I	282, 28	80–110	80–535	14, 10, 8	163	17	88	6
16	dL,I	298, 32	20–120	80–450	10, 10, 8	159	38	22	12
17	D,I	290, 09	20–130	80–500	12, 10, 10	160	23	128	4
18	O,I	290, 09	20–120	80–560	14, 9, 7	165	27	355	3
19	O,I	336, 18	20–110	80–500	15, 10, 10	156	13	637	5
20	O,Y	020, 10*	20–120	80–500	16, 10, 10	147	35	83	5
21	O,Y,bxt	– , –	20–130	80–500	16, 10, 10	142	33	10	16
22	O,Y,bxt	– , –	20–125	80–500	17, 9, 9	141	33	63	7
23	O,Y	020, 10*	20–120	80–450	13, 10, 8	143	30	124	5
24	O,Y	155, 10*	20–120	80–500	8, 8, 8	154	16	122	5
25	dL,Y	136, 18	20–120	80–500	13, 10, 8	150	44	82	6
26	O,Y	155, 10*	20–130	80–500	11, 9, 8	138	38	24	12
27	D,Y	172, 18	20–120	80–500	16, 10, 8	159	36	57	7
28	O,Y	170, 36	20–120	80–600	12, 10, 6	175	9	123	6
29	O,Y	150, 18	20–100	80–450	15, 10, 6	146	35	50	10
30	O,C	347, 15	20–120	80–500	16, 10, 7	151	29	129	5
31	L,C	347, 15	20–120	80–500	11, 10, 8	137	57	43	9
32	O,C	030, 12	20–125	80–600	17, 10, 10	145	29	53	7
33	L,C	060, 12	20–120	80–450	12, 10, 5	139	61	100	7
34	O,C	042, 15	20–135	80–535	16, 10, 7	157	25	380	3
35	D,C	042, 15	20–135	80–535	12, 10, 6	147	36	540	3
36	O,C	020, 10*	20–125	80–560	10, 10, 9	142	22	30	10
37	O,C	020, 10*	20–120	80–500	13, 10, 10	151	30	113	5

Samples: N—New Market mine; I—Immel mine; Y—Young mine; C—Coy mine; O—ore; L—limestone; dL—dolomitic limestone; D—dolostone; bxt—breccia test. Strike (Str) and Dip are given using the right-hand convention. Breccia clasts each have an attitude. Asterisks (*) indicate that the attitude comes from mine maps. Demagnetisation: range on alternating field (AF) in millitesla and on thermal (TM) step demagnetisation in degrees centigrade over which the ChRM was isolated. N, N_d, N_c—number of NRM measurements made, number of specimens demagnetised, and number of specimens used in the site mean ChRM calculation. Mean ChRM directions given by: Dec—declination; Inc—inclination; k—precision parameter; and α_{95}—95% cone of confidence.

or width, and are composed of rubble breccia with clasts from the host limestone units. The Coy ore body includes ‘‘bedded’’ breccias.

Two theories are commonly proposed for formation of the breccias. The ‘‘paleoaquifier’’ hypothesis envisages that post-Knox uplift and erosion led to the formation of the post-Knox unconformity, to the dissolution of limestone preferentially along vertical fractures, and to the formation of the solution-collapse caverns with concomitant thinning of the adjacent

Kingsport limestone by more than 50% throughout the M-JC district (Wedow and Marie, 1965; Hoagland et al., 1965; Wedow, 1971; Haynes and Kesler, 1994). However, none of the breccia bodies appear to extend up to the unconformity (Misra and Fulweiler, 1992). The paleoaquifier hypothesis implies a maximum age of latest Early Ordovician for the mineralisation. The second hypothesis suggests that the hydrothermal solutions travelled along fractures and into adjacent limestone horizons and dissolved the limestones to create caverns, leading to solution-collapse brecciation. This was followed by continued coeval dissolution of limestone and dolostone, and precipitation of dolomite and sulfides by the ore-forming brines, resulting in more than one generation of brecciation and of mineralisation because some clasts are mineralised (Hill et al., 1971a,b; Hoagland, 1976; Churnet and Misra, 1983; Taylor et al., 1983; Haynes et al., 1989; Matlock and Misra, 1993). The hydrothermal hypothesis per se implies no age constraints on MVT mineralisation. In addition, evidence for some minor sedimentary-diagenetic brecciation has been cited for the bedded breccia bodies (Churnet, 1979; Moss, 1982).

2.5. Mineralisation

The MVT mineralisation is mostly epigenetic. It fills the spaces around clasts and cements them in the breccia bodies. Sphalerite, the main ore mineral, is typically honey yellow to light brown with <1 wt.% Fe, although minor amounts of dark sphalerite with up to 5 wt.% Fe have been seen. Microscopic color banding is common (Churnet, 1979; Craig et al., 1983). Sparry dolomite, white to locally pinkish, is the main gangue mineral (Oder and Ricketts, 1961; Churnet and Misra, 1983). Ebers and Kopp (1979) found six major cathodoluminescent growth zones in the sparry dolomite that are time-correlative across the M-JC MVT district with three zones being correlative with sphalerite precipitation. In addition, the MVT mineralisation contains ubiquitous but minor amounts of pyrite and/or marcasite and silica as well as trace amounts of galena, chalcopyrite, fluorite, calcite, quartz, gypsum, anhydrite and bitumen. The distribution of sphalerite is irregular. The best concentrations are typically close to contacts of the breccia bodies with limestone host rocks. Thus, the large breakthrough breccia bodies typically have barren central cores and some are entirely barren. Typical M-JC ore has graded 2 to 4 wt.% Zn and crushed stone has also been an important by-product.

2.6. Age of sphalerite mineralisation

Prior to 1960s, the breccia bodes were thought to be tectonic and the ore to be epithermal and either syntectonic or post-tectonic (e.g., Secrist, 1924; Brokaw and Jones, 1945); however, no magmatic activity is known in the area to correlate with the mineralisation. Kendall (1960), subsequently confirmed in detail by Matlock and Misra (1993), observed that internal carbonate sediments with detrital sphalerite grains in the breccia bodies contained laminations that paralleled the host rock attitudes leading him to conclude that the mineralisation predated Alleghanian deformation. Accordingly, Kendall (1960) favored a submarine syngenetic or diagenetic origin for the mineralisation. Given that most of the mineralisation is epigenetic breccia-fill, a post-Early Ordovician age is much more likely (Churnet and Misra, 1983).

The absence of evident structural control on the mineralisation (Crawford and Hoagland, 1968; Hill et al., 1971a), the observed internal deformation of the sphalerite crystals (Taylor et al., 1983), and the offset and tilt of the bedded ores by thrust faulting and associated deformation (Fulweiler and McDougal, 1971; Hill et al., 1971b; McCormick et al., 1971), all imply that mineralisation predates either the end of or peak of late Paleozoic Alleghanian deformation.

The occurrence of the M-JC ore below the Knox unconformity led Hoagland et al. (1965) and Crawford and Hoagland (1968) to favor a pre-Middle Ordovician age for the MVT mineralisation. Hill and Wedow (1971), noting minor MVT mineralisation in Chickamauga Group above the unconformity, suggested an early Middle Ordovician age for it.

Indirect radiometric methods have attempted to bracket the age of the sphalerite mineralisation. Hearn et al. (1987) obtained a $^{40}Ar/^{39}Ar$ plateau age of 322 Ma for authigenic K-feldspar from unmineralised carbonate of the Conasauga Group in the M-JC district. They observed that K-feldspar with similar fluid inclusion temperatures and salinities to the sphalerite occurred interlocked with it in three M-JC mines. Based on the microscopic contact relationships, they

concluded that the K-feldspar predates the ore. Elliot and Aronson (1987) determined K–Ar illite–smectite ages for Middle Ordovician bentonites throughout the southern Appalachian basin, including nine ages from ~ 20 km north of the M-JC district in the Copper Ridge district that yielded an average of 291 ± 9 Ma. They concluded that fluid flow during the Alleghanian orogeny was responsible for both illitisation and MVT mineralisation, noting that Sedivy et al. (1984) had used the K–Ar method to date metamorphism by Alleghanian thrusting of the underlying Conasauga shales at 300 ± 20 Ma. Kesler et al. (1988), using Sr isotopic ratios for fluid inclusions in host rock, ore and gangue minerals, constructed a model for the thermal evolution history of brines from the adjacent Sevier shale basin, leading to an estimated 408 to 320 Ma age range for the M-JC ores.

Using the Rb–Sr sphalerite isochron method, Nakai et al. (1990) obtained a direct date of 377 ± 29 Ma (2σ) for sphalerite from the Coy mine (Fig. 3). Later, Nakai et al. (1993) reported a second Rb–Sr age of 347 ± 20 Ma for a sphalerite from the Immel Mine. Two points should be made. First, these two dates are significantly different at >95% confidence level, implying either two mineralizing events or unreliability of the method. Second, the ages correlate with the Acadian orogeny from about 380 to 350 Ma, implying that fluids derived from Acadian mountain building formed the M-JC ores. However, although Acadian orogenesis is well developed in the northern Appalachian orogen, it is not evident in the southern Appalachian Valley and Ridge Province (Misra and Fulweiler, 1992).

Finally, Bachtadse et al. (1987) completed a paleomagnetic study of specimens from the M-JC, Copper Ridge, and Sweetwater MVT district and from some intervening sites (Fig. 2). They obtained a mid-Pennsylvanian to Early Permian or 286 ± 20 Ma pole position. Accepting Kendall's (1960) argument that the mineralisation must predate Alleghanian deformation, they concluded that their synfolding remanence was formed after the MVT mineralisation. Symons et al. (1996), based on experience derived from paleomagnetic studies of several other MVT districts, decried Kendall's (1960) arguments and considered that Bachtadse et al. (1987) had dated the MVT mineralizing event in eastern Tennessee.

2.7. Thermal evidence on genesis

The thermal history of the M-JC district is quite well constrained. Conodonts yield color alteration index values of 2 to 3 (Harris et al., 1978) that are indicative of maximum temperatures of 60–140 to 110–200 °C, respectively, for the host rocks (Rejebian et al., 1987). Also, Montanez (1995) has reported fluid inclusion homogenisation temperatures from 80 to 165 °C for the regional "late-diagenetic dolomites" or dolostones from a few kilometers west of the M-JC district. The M-JC sphalerite has yielded fluid inclusion temperatures ranging from 80 to 200 °C with >95% of the values uniformly distributed between 100 to 180 °C (Taylor et al., 1983). These are consistent with the fluid inclusion temperatures of 80 to 200 °C reported by Montanez (1995) for the dolomite cements. Following Sangster et al. (1994), these results indicate a Group 1 MVT deposit that is in thermal equilibrium with matching ore and peak host rock temperatures, suggesting that the ores were deposited essentially within their own aquifer. Assuming a geothermal gradient of 30 °C/km for the district (Montanez, 1994; Steinhauff et al., 1999), this implies a burial depth of 3 to 6 km during sphalerite deposition. From the Thornhill section about 15 km north of the M-JC district, the thickness of the post-Knox unconformity stratigraphic sequence, spanning from Middle Ordovician to lower Mississippian, is ~ 1.80 km thick (Walker and Lutz, 1992). Thus, the paleotemperature data requires either, more probably, burial by thrust fault repetition of the Paleozoic stratigraphic sequence by 1–4 km or, less probably, the entire region being quite uniformly dolomitised and mineralised by hydrothermal fluids migrating along aquifers and saturating and heating the adjacent strata.

3. Experimental methods and results

3.1. Sampling

All 37 sampling sites for this study were collected from the Coy (8 sites), Immel (10), New Market (9) and Young (10) mines of ASARCO in the M-JC district (Fig. 3), in the upper third of the Kingsport and lower fifth of the Mascot formations. The site locations were selected to be broadly representative of

the different lithologies present, i.e., limestones, dolostones and ore mineralisation from both breakthrough and bedded breccia bodies in several units (Table 2). They included three sites of breccia clasts to provide a paleomagnetic breccia test and four sites of maximum dip away from a dome to provide a paleomagnetic fold test. The sites were also chosen to provide geographic coverage of the M-JC district, of each mine, and of two thrust sheets (Fig. 3). Collecting in the mines is advantageous because the rocks are unweathered so that modern secondary hematite or goethite is not present to possibly confuse remanence analysis.

At each site, four or more block samples were collected and oriented using a Brunton compass. From each block sample, three or more oriented core specimens were drilled, each with a 2.54 cm diameter and 2.20 cm length. The 576 specimens were then stored for more than 2 months in a magnetically shielded room, which has an ambient field of $\sim 100\gamma$, to allow their viscous remanent magnetisation (VRM) components to decay. All subsequent measurements were done without removing the specimens from the room.

3.2. Natural remanent magnetisation

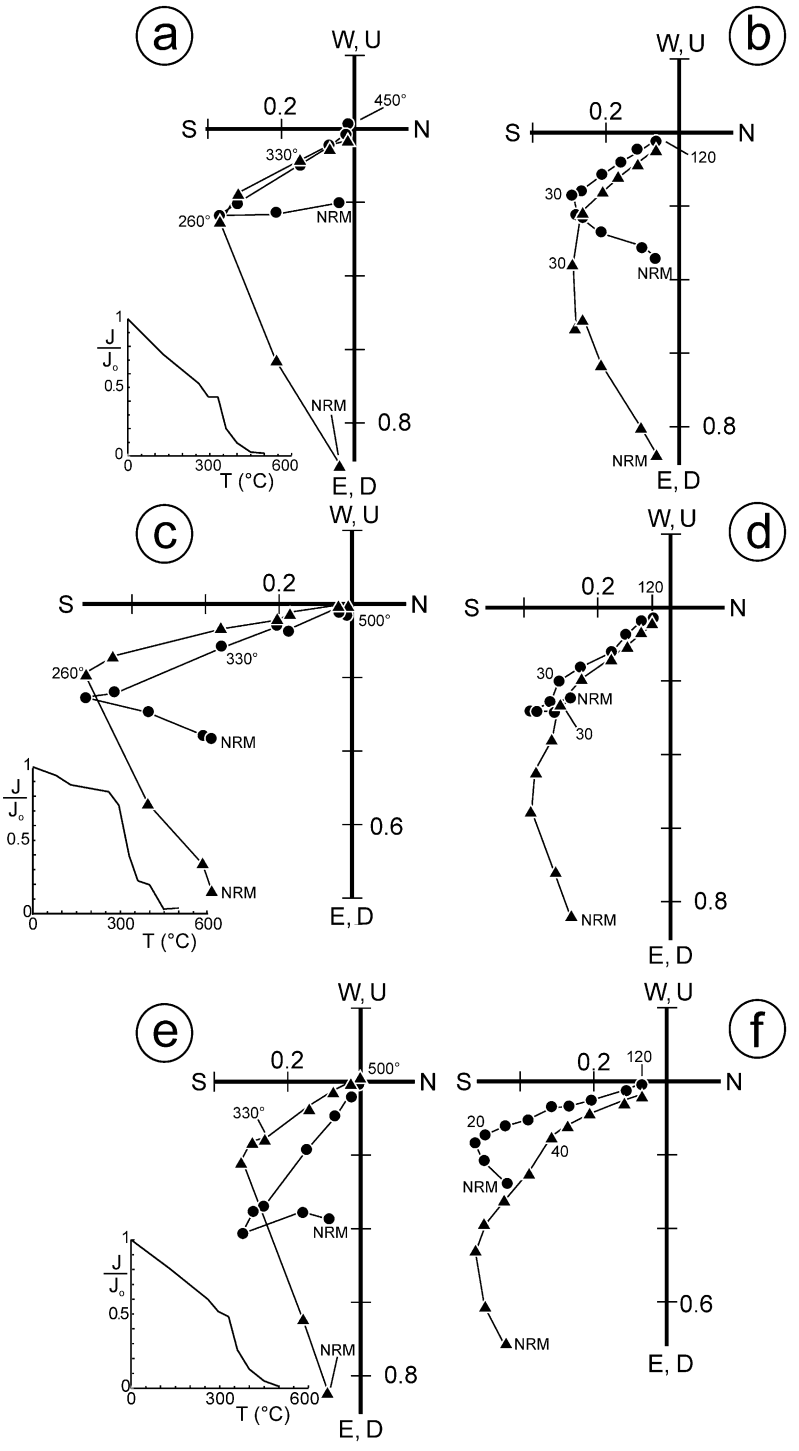
The natural remanent magnetisation (NRM) of each specimen was measured using an automated Canadian Thin Films DRM-420 cryogenic magnetometer with a resolution limit of $\sim 3 \times 10^{-6}$ A/m for a specimen's remanence intensity. Grouping the data, the specimens from the 8 limestone and dolomitic limestone sites, 3 dolostone sites and 26 ore sites yielded median NRM intensities of 4.2 (first quartile, $Q_1 = 3.3$, third quartile, $Q_3 = 6.3) \times 10^{-4}$, 12.7×10^{-4}, and 7.5 ($Q_1 = 4.8$, $Q_3 = 12.2) \times 10^{-4}$ A/m, respectively. The salient point is that MTV mineralisation is about twice as intensely magnetised as barren limestone and dolomitic limestone host rocks, implying that the remanence was acquired during the mineralising event. Also, the NRM intensities of the M-JC ores and host rocks are comparable to those measured to date for MVT ores elsewhere (Symons et al., 1996). From the NRM data, about 10 specimens with the most intense NRMs were selected for each site, requiring that each of the four blocks be represented by at least two specimens. This process reduced the collection to 369 specimens for paleomagnetic analysis (Table 2).

3.3. Thermal step demagnetisation

One specimen from each of 36 sites and two specimens from one breccia sites were thermally demagnetised in 12 steps from 80 to 600 °C using a Magnetic Measurements TD-1 oven. After initially removing the residual VRM, virtually all of the limestone, dolostone and ore breccia specimens showed isolation in the $\sim 265-550$ °C range of a characteristic remanent magnetisation (ChRM) component that is directed southeasterly and downward at an intermediate angle (Fig. 4a,c,e). Specimens of ore mineralisation, including those of $\sim 60 \pm 10\%$ sphalerite, gave the same response (Fig. 5a,c).

The presence of distinct ChRM intensity decreases preferentially in the 275 to 325 °C range indicates that pyrrhotite is a significant ChRM carrier, and the continued presence of the ChRM above 325 to almost 585 °C indicates that magnetite is also a carrier (Fig. 4a,c). The relative importance of the two minerals as carriers varies in mineralised specimens, but both pyrrhotite and magnetite appear to be present in most specimens with the pyrrhotite contributing a little more to their ChRM. Thermal demagnetisation indicates that only magnetite is present in the limestone sites, that magnetite is more abundant than pyrrhotite in the dolomitic limestone sites, and that the converse is true for the dolostone sites.

Fig. 4. Vector component diagrams showing step demagnetisation of example specimens of: (a, b) limestone; (c, d) dolostone; and (e, f) breccia clasts. The axes for the horizontal plane are north (N), east (E), south (S) and west (W), and for the vertical plane are up (U), north (N), down (D) and south (S), and they are expressed as the ratio of the NRM intensity (J_o) equal to 1.0. The specimen numbers and the J_o values in A/m ($\times 10^{-4}$) are: (a) 250103, 7.46; (b) 250204, 7.01; (c) 170403, 17.2; (d) 350401, 37.2; (e) 220101, 53.7; and (f) 210201, 12.6. Vector points are represented by circles and triangles for the horizontal and vertical planes, respectively. Panels a, c and e show thermal data with some steps labelled in °C, as well as a plot of the decrease in NRM intensity (J/J_o) against temperature (°C) showing a marked drop in the pyrrhotite unblocking temperature range of ~ 265 to 330 °C (panels a, c) as well as in the magnetite range thereafter or in magnetite only (panel e). Panels b, d and f show alternating field data with some steps labeled in mT.


a
W, U
0.2
450°
S
N
330°
260°
NRM
NRM
0.8
E, D
J/J0
1
0.5
0
0
300
600
T (°C)
b
W, U
0.2
S
N
120
30
30
NRM
NRM
0.8
E, D
c
W, U
0.2
S
N
500°
330°
260°
NRM
NRM
0.6
E, D
J/J0
1
0.5
0
0
300
600
T (°C)
d
W, U
0.2
120
S
N
30
NRM
30
NRM
0.8
E, D
e
W, U
0.2
500°
S
N
330°
NRM
NRM
0.8
E, D
J/J0
1
0.5
0
0
300
600
T (°C)
f
W, U
0.2
120
S
N
20
40
NRM
NRM
0.6
E, D

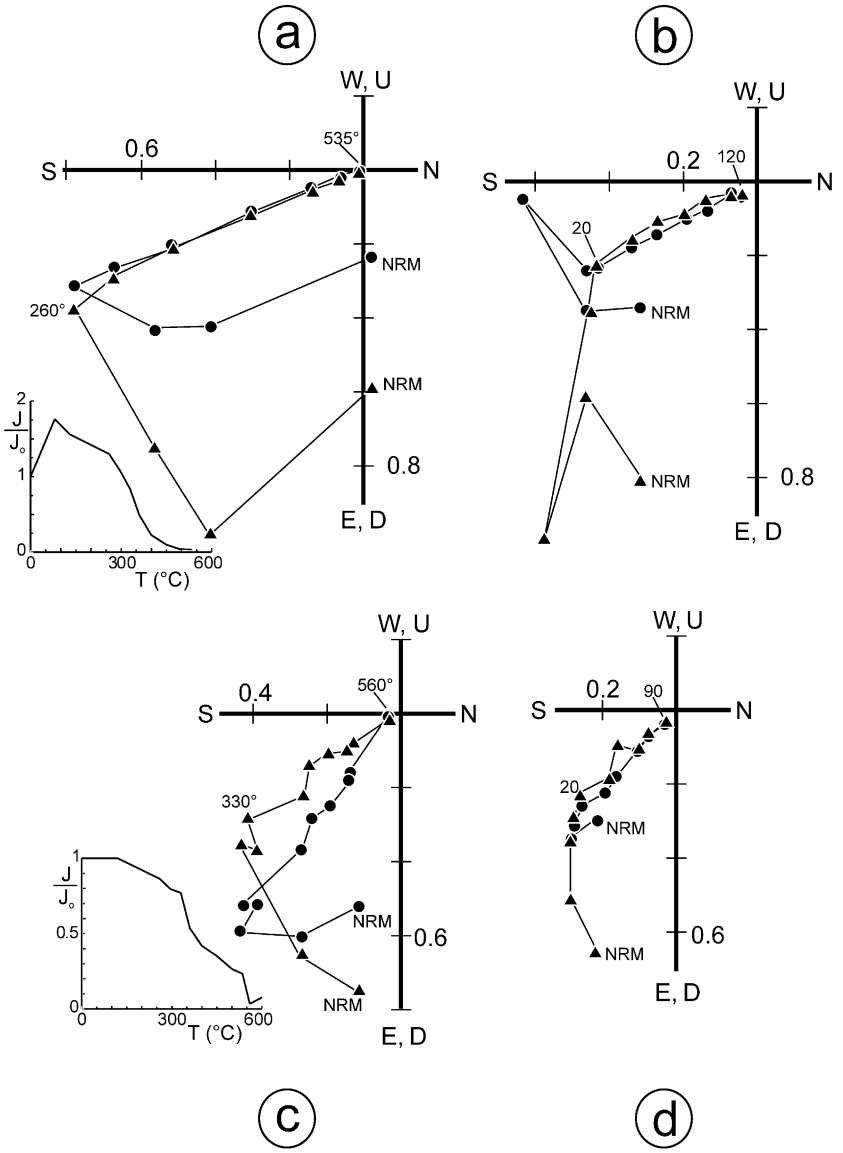


Fig. 5. Vector component plots of example specimens of ore with $60 \pm 10\%$ sphalerite on thermal (a, c) and alternating field (b, d) step demagnetisation. Conventions as in Fig. 4. The specimen numbers and the J_o values in A/m ($\times 10^{-4}$) are: (a) 060404, 10.0; (b) 030202, 14.5; (c) 320202, 19.2; and (d) 230301, 2.3.

The ChRM directions were identified using visual inspection of orthogonal demagnetisation diagrams (Zijderveld, 1967) and calculated using the least-squares fitting method of Kirschvink (1980), requiring for acceptance that three or more demagnetisation steps be included and that the specimen's mean angular deviation be $<10°$.

3.4. Alternating field step demagnetisation

Two specimens from each site, or 74 in total, were demagnetised in an alternating field (AF) in 14 steps of 5 to 15 mT increments up to 130 mT using a Sapphire Instruments SI-4 demagnetiser. Again, virtually all specimens isolated a ChRM component to the south-

east and downward at an intermediate angle between ~ 20 and ~ 120 mT after initially removing the residual VRM (Fig. 4b,d,f), including the predominantly sphalerite specimens (Fig. 5b,d). Upon noting that AF and thermal step demagnetisation gave similar ChRM directions, the remaining specimens selected from each site were AF demagnetised in 5 to 7 steps between ~ 35 and ~ 100 mT to isolate their ChRM (Table 2).

3.5. Saturation isothermal remanent magnetisation

Fourteen additional virgin specimens, representing the various lithologies and mines, were selected for saturation isothermal remanent magnetisation (SIRM) testing to examine the domain size distribution of their magnetic minerals. After initial AF demagnetisation to 130 mT, these specimens were progressively pulse magnetised in a Sapphire Instruments SI-6 pulse magnetiser in eight direct field steps up to 900 mT and then AF demagnetised in seven steps to 130 mT. All specimens show rapid SIRM acquisition and demagnetisation curves that are typical of both pyrrhotite and magnetite (Fig. 6); however, they provide a better fit to pyrrhotite type curves on a crossover plot, confirming that it is the predominant carrier (Symons and Cioppa, 2000). The plot also shows that the carriers are pseudosingle domain in size and that they are not interacting magnetically (Fig. 6). Also, the coercivities at the crossover points of 35 to 60 mT are typical of pyrrhotite and distinctly greater than the comparable 15 to 35 mT values typical of single to pseudosingle domain magnetite. Note that the domain sizes of the sphalerite-rich specimens are slightly larger that those of most mineralisation and the "barren" host rocks. This analysis indicates that the ChRM of all specimens would be magnetically highly stable over geologic time intervals of $\gg 10^6$ years.

3.6. Dolomite/sphalerite magnetisation

After excluding chipped or glued specimens, the specific gravities of the MVT ore specimens were determined by dividing their volume into their weight, and the weight percentages of sphalerite were determined by assuming reasonably that only dolomite and sphalerite were present. The NRM intensities for the 41 lowest grade specimens (< 2 wt.% sphalerite) were then compared to those for the 42 highest grade specimens (50–70 wt.% sphalerite) and the median intensity was found to increase from 6.63 ($Q_1 = 4.50$, $Q_3 = 9.40$) $\times 10^{-4}$ to 8.45 ($Q_1 = 4.91$, $Q_3 = 14.50$) $\times 10^{-4}$ A/m. This result indicates that the pseudosingle domain pyrrhotite and magnetite that carry the ChRM occurs in both the dolomite and sphalerite of the ore with the latter carrying a somewhat greater percentage.

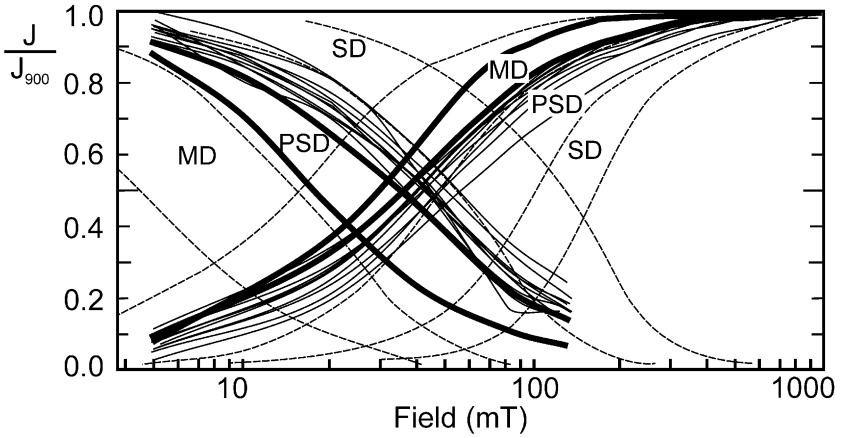


Fig. 6. Crossover plot showing the acquisition of saturation isothermal remanent magnetisation (SIRM) as a ratio of the intensity at 900 mT in a direct field (J_{900}) as increasing curves and the alternating field demagnetisation of the SIRM to 120 mT as decreasing curves (solid lines) relative to type curves for single, pseudosingle and multidomain (SD, PSD, MD) pyrrhotite (dashed lines, Symons and Cioppa, 2000). The fine solid lines are for all lithologies except two specimens (heavy solid line) of 60 ± 10% sphalerite ore.

This result further indicates that the ChRM is coeval with the mineralisation stage of the paragenetic sequence.

4. Data analysis

4.1. Site mean remanence directions

Sites 1 and 11 did not give a coherent cluster of ChRM directions for five or more specimens from at least three block samples and were rejected from further analysis. The remaining 35 sites all gave well-clustered directions, utilising data from >86% of the step demagnetised specimens. Their site mean directions were calculated from the ChRM directions following Fisher (1953) (Table 2, Fig. 7) and yielded a median radius for their cones of 95% confidence of 6° ($Q_1 = 5°$, $Q_3 = 7°$) that indicates a high degree of within-site consistency.

4.2. Breccia test

The paleomagnetic breccia test is a variant of the conglomerate test (Graham, 1949). If breccia clasts have random ChRM directions, then they have not been remagnetised since the breccia formed, and it is likely that adjacent similar rocks have not been remagnetised either. Conversely, if the clasts have similar ChRM directions, then their ChRM is a post-brecciation remagnetisation and similar adjacent lithologies are likely to be remagnetised also. The 17, 10 and 9 clasts from sites 4, 21 and 22 give precision parameter (k) values of 53, 10 and 63, respectively. These k values are very much greater than the comparison statistic of $k = 1.54$ to 1.89 for population of these sizes, indicating with $\gg$95% confidence that the ChRM directions form non-random or clustered populations for all three tests, meaning that all of the clasts have been thoroughly remagnetised (Watson, 1956). Further, the three breccia clast sites give site mean directions that are indistinguishable from those of the MVT ore sites (Fig. 7a), indicating that the clasts were remagnetised when the ore was formed.

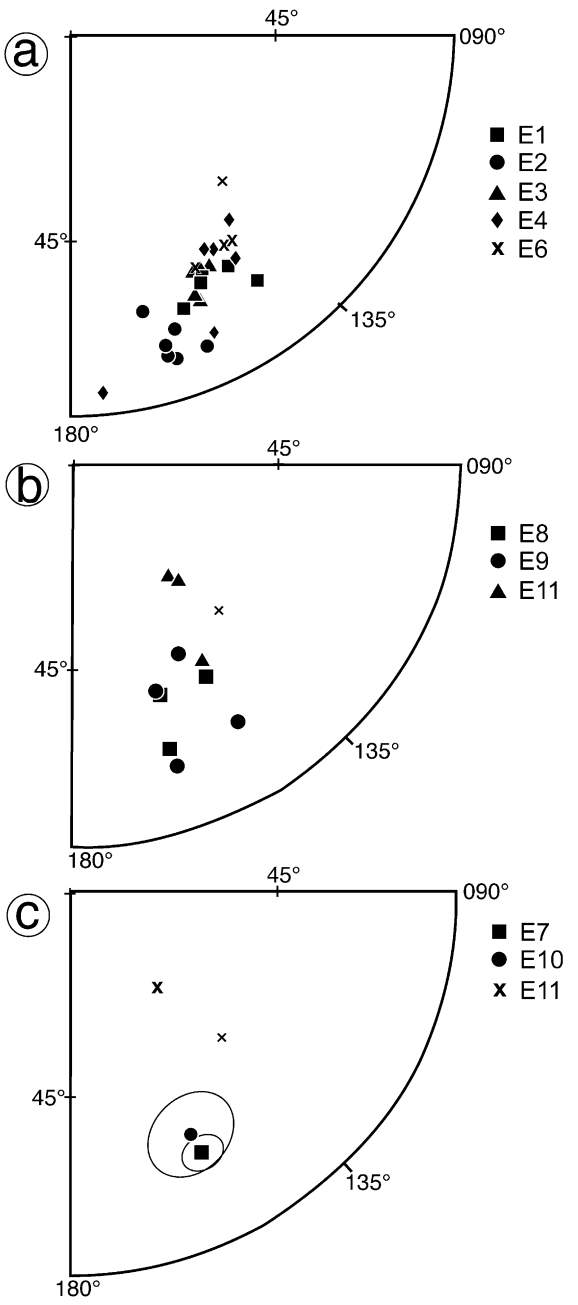


Fig. 7. Equal-area stereograms showing the southeast-lower quadrant with the: (a) site mean ChRM directions for MVT mineralisation in the Coy (square), Immel (circles), New Market (triangles) and Young (diamonds) mines, and in breccia clasts (X's); (b) site mean ChRM directions for host rock dolostones (squares), dolomitic limestones (circles) and limestones (triangles); and (c) unit mean ChRM directions with their cones of 95% confidence for all MVT mineralisation (square) and dolostone and dolomitic limestones host rocks, and the mean only for the limestone host rocks (X).

4.3. Fold test

The paleomagnetic fold test asserts that the ChRM predates folding if the degree of clustering of the ChRM directions improves significantly and postdates folding if the clustering degrades significantly on tilt correction (Graham, 1949). Several fold tests were run. Most dips are relatively gentle so that the fold tests are not highly resolved. For some mineralised site, the bedding attitudes were derived from adjacent strata as the original bedding was not longer visible (Table 2). The McElhinny (1964) test and the McFadden and Jones (1981) test using 10° unfolding increments were run for the mineralised sites, excluding breccia clast sites, for each mine separately. For the New Market, Immel, Young and Coy mines, the best grouping was found at 20%, 20%, 0% and 0% unfolding, respectively, with the resulting clusters being statistically significant at 95% confidence. These results imply that folding was 90 ± 10% completed at the time of magnetisation. Further, the best single structure to test is the broad domal structure in the Young Mine using specimens from sites 20 and 23 on its eastern side and sites 27 and 28 on its western side. Again, the best grouping is at 0% unfolding, and the result is significant at 95% confidence for both fold tests.

The fold tests showed that the eight dolomitic limestone and dolomite sites are best grouped at 30% unfolding and form a significantly improved cluster at 95% confidence, and that the three limestone sites are best clustered at 70% unfolding but not significantly better at 95% confidence. The implication is that the limestones retain a substantial prefolding component in the ChRM that is being removed by dolomitisation and that it is entirely lost during mineralisation.

To examine the effect of post-mineralisation faulting on tilt, it is instructive to deem each mine to be a single fold limb using the average ChRM direction and attitude for its mineralised sites, and giving a weighting to each limb that is equivalent to its number of sites. When this is done, the fold tests show that the best clustering is at 0% untilting where the directions form a statistically significant population, and untilting causes progressive degradation to 100% untilting with non-significance reached at ~ 45% untilting. Further, because the dips of the strata average to within 4.0 ± 6.6° of horizontal, the overall direction on 100% untilting changes by only 2.4° of arc. Therefore, even if a small contribution of synfolding remanence (e.g., 20% untilting) is present, it will not move the overall ChRM direction or pole position by even 1° of arc.

4.4. Paleomagnetic poles

There are five or six accepted site mean directions for MVT mineralisation in each of the four M-JC mines. When the 22 site mean directions are plotted on a stereonet, they form a single population to the south-southeast at a moderately downward inclination (Fig. 7a). Each mine mean direction without tilt correction is statistically indistinguishable from one or more of the other mine means (Fisher, 1953; Table 3), indicating that they can be combined to form a single population (McFadden and Lowes, 1981). This also indicates that there has not been a significant relative rotation between the thrust sheets above and below the

Table 3
Unit mean remanence directions for the M-JC district

Entry	Unit	Sites	Mean remanence direction				
			N	Declination (°)	Inclination (°)	α_{95} (°)	*k*
E1	Coy—ores	30, 32, 34, 36, 37	5	149.2	27.1	5.9	168
E2	Immel—ores	10, 12, 14, 15, 18, 19	6	161.4	17.3	5.2	169
E3	New Market—ores	3, 6–9	5	152.0	28.6	3.6	445
E4	Young—ores	20, 23, 24, 26, 28, 29	6	151.3	27.8	14.1	23
E5	Ores	E1–E4	22	153.9	25.0	4.4	51
E6	Breccia clasts	4, 21, 22	3	145.0	32.4	8.1	235
E7	MVT mineralisation	E5+E6	25	152.9	26.0	4.1	51
E8	Dolostone	17, 27, 35	3	155.5	31.8	14.8	70
E9	Dolomitic limestones	2, 13, 16, 25	4	153.8	30.6	15.9	34
E10	Dolomitic host rock	E8+E9	7	154.6	31.1	8.5	52
E11	Limestones	5, 31, 33	3	141.4	52.4	18.6	45

Rocky Valley thrust fault since the MVT mineralisation event (Fig. 3). Note also that the three sites of breccia clasts give a mean direction that is indistinguishable from the MVT mineralisation (Fig. 7a; Table 3). Combining all 25 site means gives a unit mean direction of Dec = 159.2°, Inc = 26.0°, α_{95} = 4.1°, and k = 51 (Table 3, entry 7), which gives a pole position at longitude 128.7°E, latitude 34.0°N with semi-axes of its oval of 95% confidence of d_p = 2.4° and d_m = 4.4° (Fig. 8).

Ten sites of host rock specimens gave accepted site mean directions. Of these, two limestone sites (#31, 33) gave very much steeper inclinations than any of the others (Fig. 7b). The remaining five limestone sites gave directions that are statistically the same as the three dolostone sites. Staining with Alizarin Red and potassium ferrocyanide showed that the two aberrant sites plus one other limestone site (#5) were undolomitised whereas four of the five coherent limestone sites were dolomitic, as were the three dolostone sites. Therefore, the seven dolomitic limestone and dolostone site mean directions were combined to give a unit mean direction of Dec = 154.6°, Inc = 31.1°, α_{95} = 8.5° and k = 52 (Table 3), which gives a pole position of 125.3°E, 31.9°N (d_p = 5.3°, d_m = 9.4°). The three true limestone sites, applying a 70% tilt correction, give a unit mean direction of Dec = 137.2°, Inc = 52.2°, α_{95} = 5.9°, k = 131 with a pole position of 131.1°E, 10.3°N (d_p = 11.0°, d_m = 16.2°), which falls closest to the late Ordovician–Early Silurian portion of the apparent polar wander path (APWP) for the North American craton from Van der Voo (1993) (Fig. 8a).

The cone of 95% confidence about the mean ChRM for the dolomitised host rocks entirely sur-

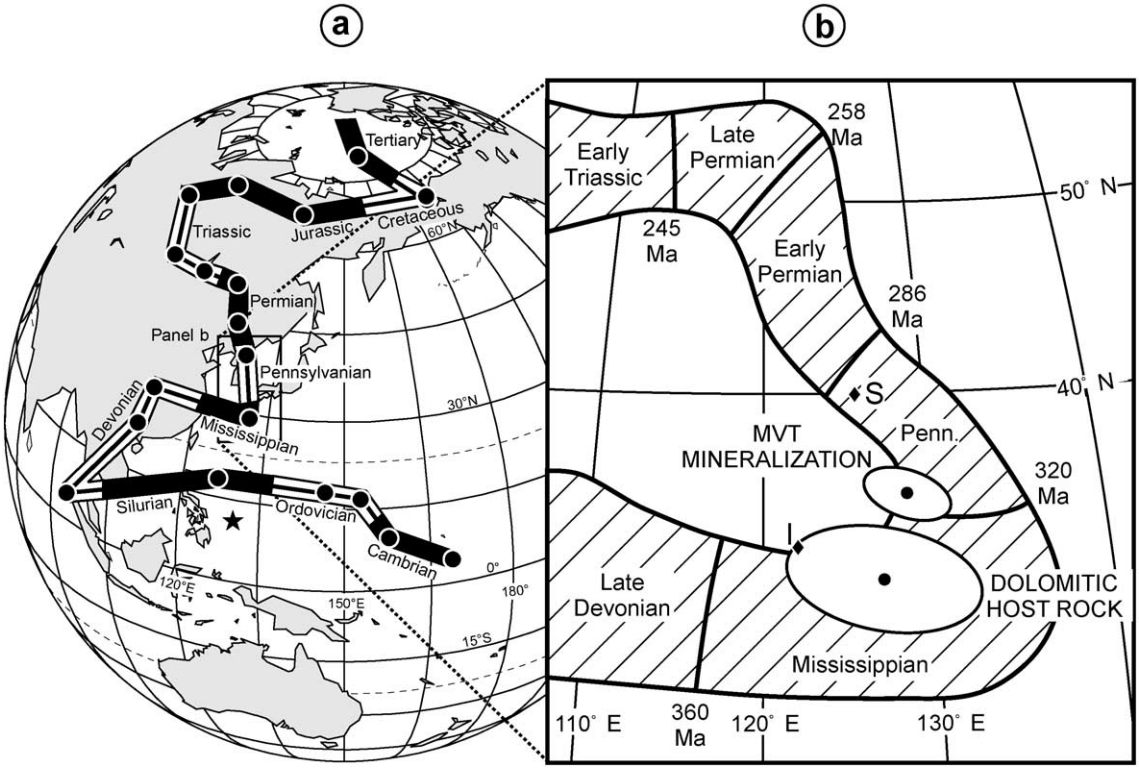


Fig. 8. Apparent polar wander path for North America from Besse and Courtillot (1991) and Van der Voo (1993) from: (a) the Paleozoic to Present with the location of the pole (star) for the three limestones sites; and (b) the Late Devonian to Early Triassic in detail with the poles for the M-JC MVT mineralisation and dolomotic host rocks with their 1σ ovals of confidence, and the in situ (I) and synfolding (S) poles from Bachtadse et al. (1987) that have 1σ ovals comparable in size to the dolomitic host rock's oval.

rounds the cone for the MVT mineralisation, showing without further testing that the two directions are not significantly different at the 95% confidence (2σ) level. However, when these poles are plotted on the APWP for North America, their relative positions are instructive (Fig. 8b). In Fig. 8b, the APWP is shown with its 1σ confidence limits, as are the MVT mineralisation and dolomitic host rock poles. In general, overlap of the 1σ limits shows contemporeinity at >95% confidence. Scaling time along the APWP, the determined age for the MVT mineralisation is 316 ± 8 Ma (1σ) and for the dolomitisation is 334 ± 14 Ma (1σ). The error on the age was obtained conservatively by noting the maximum age difference defined by the error oval.

5. Discussion

5.1. Comparison to Bachtadse et al.'s (1987) paleomagnetic results

Bachtadse et al. (1987) measured ~200 specimens paleomagnetically from ~55 block samples from seven surface sites in the Copper Ridge district (2), Sweetwater district (3) and other regional sites (2) (Fig. 2) and one underground site in the New Market mine (Fig. 3). At each site, they sampled a variety of limestones, dolostones, breccia clasts, dolomites and sphalerite mineralisation. Finding that all lithologies gave similar NRM intensities and ChRM directions, they opted to report site mean ChRM directions only. Their eight site means are loosely clustered with a high average α_{95} value of 18.4°, likely reflecting, in part, the within-site lithologic variation. When the site means are averaged to get a unit mean, their results are similar to ours but not directly comparable because of the contrast in geographic distribution and treatment of lithologies.

One important difference in the studies is that Bachtadse et al. (1987) reported only magnetite as a ChRM carrier except for one site with secondary hematite. In contrast, we report both pyrrhotite and magnetite as coeval carries of about equal intensity. The presence of a sulfide-borne remanence is consistent with the notion that ChRM is contemporaneous with sphalerite deposition during the MVT mineralisation rather than a post-mineralisation event as Bachtadse et al. (1987) suggested.

As we found, Bachtadse et al. (1987) reported that breccia clasts at some sites, excluding one site in the Sweetwater district, gave clustered ChRM directions to indicate a post-brecciation ChRM. They ran a fold test also. Although it that was not statistically significant at 95% confidence, the best clustering of the ChRM occurred at 60% tilt correction, leading them to favor a synfolding ChRM that was acquired during the Alleghanian orogeny. Their test is based on eight site mean directions from a large area that utilises diverse lithologies and structures. Drawing on our observation that the regional limestones appear to carry an early Paleozoic and predominantly prefolding ChRM, it is likely that their limestone samples are contributing to the large within-site scatter at some of their sites and to their synfolding test result.

Bachtadse et al. (1987) determined an in situ pole position of 32.8°N, 121.7°E ($d_p = 6.3°$, $d_m = 11.2°$) and a synfolding pole of 39.9°N, 126.3°E ($d_p = 4.7°$, $d_m = 8.5°$). They considered the fully tilt-corrected or prefolding pole to be invalid because it was too discordant with the APWP. Their in situ pole is similar to our dolomitisation pole (Fig. 8b) and it indicates a Mississippian age using the Van der Voo (1993) APWP. This suggests a close temporal relationship between the M-JC dolomitisation event and those in the adjacent MVT districts and intervening Knox Group strata.

5.2. Comparison to other M-JC radiometric dating evidence

The ages obtained for M-JC mineralisation and dolomitisation are shown in Fig. 9 relative to other age evidence. The 316 ± 8 Ma age for the MVT mineralisation does not fit the strontium isotope data, i.e., it is significantly younger than either of the two Rb–Sr sphalerite isochrons determined by Nakai et al. (1990, 1993), and it falls at the end of the long time interval of acceptability obtained from strontium isotope modeling (Kesler et al., 1988) and fluid inclusion chemistry (Haynes et al., 1989). On the other hand, the 316 ± 8 Ma paleomagnetic age fits closely with the argon isotope data. It agrees with the 322 Ma $^{40}Ar/^{39}Ar$ plateau age obtained by Hearn et al. (1987) from authigenic feldspar from the Conasauga Formation that is also associated with sphalerite in the M-JC district and with the K–Ar illite ages of 300 ± 20

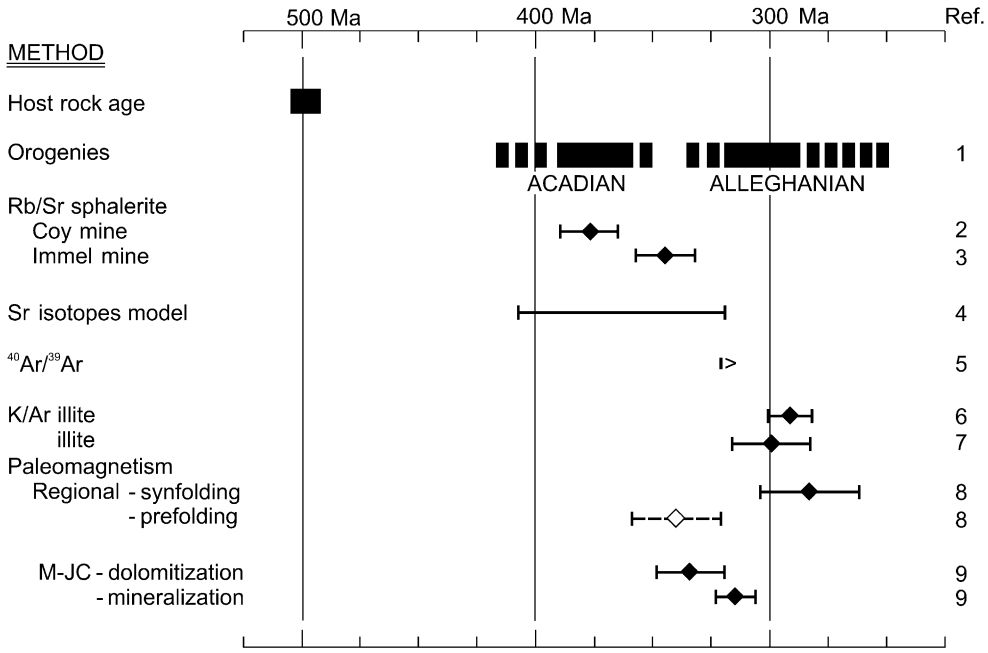


Fig. 9. Tabulation of radiometric and paleomagnetic age dating results with 1σ errors for MVT mineralisation and dolomitisation in the M-JC district. (1) Osberg et al. (1989), Garven et al. (1993), Hamilton-Smith (1993), Hatcher et al. (1989), Steinhauff et al. (1999); (2) Nakai et al. (1990); (3) Nakai et al. (1993); (4) Kesler et al. (1988); (5) Hearn et al. (1987); (6) Elliot and Aronson (1987); (7) Sedivy et al. (1984); (8) Bachtadse et al. (1987), showing their preferred synfolding pole result and alternative post-folding result; and (9) this study.

Ma from the Conasauga shales (Sedivy et al., 1984). However, the 316 ± 8 Ma M-JC ore age is older than the 291 ± 9 Ma K–Ar illite age from Ordovician bentonites in the adjacent Copper Ridge MVT district (Elliot and Aronson, 1987), implying that it and the M-JC district may not represent coeval mineralisation events but rather be sequential events.

5.3. Dolomitisation age

The 334 ± 14 Ma date from paleomagnetism for regional dolomitisation conflicts with interpretations that support an Early Ordovician syngenetic origin (Laporte, 1971; Harris, 1973; Churnet and Misra, 1981; Churnet et al., 1982; Haynes and Kesler, 1994), a Middle Ordovician karsting-related event (Kendall, 1960; Churnet and Misra, 1981), or a syn-mineralisation event (Oder and Ricketts, 1961; Haynes et al., 1989). Rather, the 334 ± 14 Ma date indicates that dolomitisation immediately preceded MVT mineralisation in the M-JC district, as Hoagland (1976) suggested and Montanez (1994) determined from the paragenetic sequence, over a period of ~ 30 Ma. Several caveats should be noted. First, this 334 ± 14 Ma date is based on relatively few sites and could be considerably improved by additional paleomagnetic analyses. Second, the date reflects the timing for the vast majority of the regional dolomitisation—minor earlier or later events could be present also. Third, the possibility cannot be entirely excluded that very fine-grained dolostones were formed earlier, and that they were simply recrystallised to coarser grain sizes at 334 ± 14 Ma by either increased burial pressures or fluids to reset their ChRM (McCabe and Elmore, 1989; Elmore et al., 1993; Lewchuk et al., 2000; Cioppa et al., 2000). Finally, the fact that dolomitisation precedes MVT mineralisation by ~ 18 Ma on average is entirely consistent with values observed for several major MVT districts throughout North America to date, including the nearby central Tennessee MVT district (Fig. 1; Lewchuk and Symons, 1995).

5.4. Comparison to geologic dating arguments

Several geologic arguments have been made concerning the age of the M-JC MVT deposits. The paleomagnetically determined age of 316 ± 8 Ma and the results of the breccia tests show unequivocally that the MVT mineralisation is epigenetic and post-dates brecciation. Note, however, the breccia could have been formed by either the Early to Middle Ordovician post-Knox paleoaquifer hypothesis or by the hydrothermal dissolution–precipitation hypothesis as described in Section 2.4. For the paleoaquifer hypothesis, our result implies that the open spaces in the breccia bodies remained voids from Early Ordovician to Mississippian time, or for >100 Ma through the time of the Taconic and Acadian orogenies. Haynes and Kesler (1994) propose such a scenario, except that they favor an Acadian orogenic age for mineralisation. For the hydrothermal dissolution–precipitation hypothesis, our result simply limits the mechanism to be a pre-Permian event and, unless there were multiple hydrothermal events, to be Mississippian. A prefolding and prefaulting genesis for the MVT mineralisation has been postulated by Kendall (1960) based on the presence of stratabound M-JC ore, by Matlock and Misra (1993) based on apparently detrital sphalerite grains in internal sediments that have the same attitude as their host rocks, by Crawford and Hoagland (1968) and Hill et al. (1971a) based on the absence of apparent structural control of the mineralisation, and by Fulweiler and McDougal (1971), Hill et al. (1971b) and McCormick et al. (1971) based on the tilting and offset of bedded ores by thrust faulting. However, all of these prefolding or prefaulting features could be formed also by hydrothermal fluids dissolving amenable carbonate lithologies to create open spaces or moulds into which epigenetic MVT mineralisation was subsequently emplaced. Further, other MVT districts have been shown to have stratabound but syn- and post-folding ores (e.g., Kicking Horse and Robb Lake, British Columbia, Symons et al., 1998a; Smethurst et al., 1999), to have stratiform "detrital" sphalerite and dolomite grains that actually post-date tilt (e.g., Polaris, Arctic Canada, and Newfoundland Zinc, Symons and Sangster, 1992; Pan and Symons, 1993), to be flat-lying undeformed and stratabound but to host much younger MVT mineralisation than the host rock age (e.g., central Tennessee and the Viburnum Trend of southeastern Missouri, Lewchuk and Symons, 1996; Symons et al., 1998a,b), and to have fault-offset mineralised strata but to be post-faulting ores (e.g., Upper Silesia, Poland, Symons et al., 1995). The observed internal deformation in the sphalerite crystals (Taylor et al., 1983) is less easily explained because it does appear to require increased post-mineralisation stress from either burial or tectonism. It should be realised, however, that a post-mineralisation planar movement on a fault will not change the remanence direction of the offset strata. The direction will change only if there is rotation on the fault or concurrent folding of the strata, i.e., considerable thrust-fault thickening and vertical jostling of the M-JC strata could have occurred without altering the determined paleomagnetic age.

5.5. Orogenic considerations

Direct age dating studies of a dozen of the world's major MVT deposits has shown a one-to-one correlation to date between the deposit's age and the age of a major orogenic event in the surrounding or adjacent orogen (Pan et al., 1990; Sangster and Symons, 1991; Symons and Sangster, 1992, Symons et al., 1996, 1998b; Smethurst et al., 1999). At first glance, the M-JC district seems to provide a poor fit to this correlation because the Acadian and Alleghanian orogenies are thought to span from ~ 390 to ~ 350 and from ~ 320 to ~ 250 Ma, respectively (Hatcher et al., 1989; Garven et al., 1993), leaving the 316 ± 8 Ma ore-forming event just straddling the start of the Alleghanian orogeny. This is awkward because the M-JC ore-forming event is shown to be post-folding yet appears to occur almost at the start of the Alleghanian orogeny.

The time span of the Acadian orogeny in the central Appalachian orogen is fairly consistently defined. In their summary article, Osberg et al. (1989) gave a span from ~ 415 to ~ 360 Ma with peak orogenesis at ~ 390 Ma based broadly on radiometric, metamorphic, structural, and other evidence. Garven et al. (1993) set the limits at ~ 395 and ~ 350 Ma for the purpose of hydrologic modeling. Hamilton-Smith (1993) leaned heavily on sedimentological evidence and placed the limits at ~ 390 and ~ 360 Ma. Thus, both 334 ± 14 Ma dolomitisation and subsequent

MVT mineralisation events post-date the Devonian Acadian orogeny (Fig. 9).

The timing of the Alleghanian orogeny is diachronous throughout the Appalachian orogen. In their extensive summary article, Hatcher et al. (1989) gave ~ 330 Ma as the start and ~ 286 to ~ 266 Ma as the end of the orogeny. Subsequently, Garven et al. (1993) accepted slightly younger limits from ~ 320 to ~ 250 Ma, and Steinhauff et al. (1999) accepted slightly broader limits from 332 to 250 Ma. For all three of these cited assessments, the paleomagnetically determined age for dolomitisation of 334 ± 14 Ma predates the starting date for the orogeny and the 316 ± 8 Ma mineralisation date is near the start of the orogeny before much deformation might be expected. Thus, a closer examination of orogenic timing is warranted.

Focusing much more narrowly on the eastern Tennessee region, the evidence is more favorable. In the Tennessee salient of the Valley and Ridge Province, the orogeny produced the Pennington–Lee clastic wedge of shallow marine deltaic sediments. The oldest sediments are probably Meramecian at ~ 333 ± 4 Ma or mid-Mississippian, the youngest are Desmoinsean at ~ 307 ± 3 Ma or mid-Pennsylvanian, and the period of most rapid deposition is Chesterian at ~ 326 ± 6 Ma or upper Mississippian (Hatcher et al., 1989). The sediments were derived from ~ 100 to ~ 200 km to the southeast in North Carolina, shed from uplifted highlands, Topographically, these highlands are thought to have risen ~ 2–3 km above the M-JC district, providing ample gradient to drive fluid flow (Kyle, 1976; Stearns and Reesman, 1986; Garven et al., 1993). From dynamic modeling, Beaumont et al. (1987) estimated the thickness of the sediments in the uplifted foreland basin to be on the order of 11 km, about triple the known stratigraphic thickness, indicating concomitant stratigraphic repetition from thrust faulting.

The timing of deformation in the Tennessee salient of the Valley and Ridge Province is imprecise. The Alleghanian "flat-on-flat" style of thrust faults in the Pine Mountain thrust sheet on the northwestern side of the province cut strata as young as the Desmoinsean Breathitt Group, indicating some displacement after ~ 307 ± 3 Ma (Hatcher et al., 1989). On the southeastern side of the province, and only a few kilometers from the M-JC district, the youngest strata cut by the frontal "ramp-and-flat" style faults of the Blue Ridge–Piedmont thrust sheet belong to the Meramecian Greasy Cove Formation, indicating displacement after ~ 333 ± 4 Ma. Accepting the inside-to-outside deformation plan of Hatcher and Odom (1980) for the Alleghanian orogeny as fundamentally correct for the region, deformation in the M-JC district could have progressed substantially prior to the 316 ± 8 Ma mineralizing event. Further, there is evidence of considerable orogenic activity to the southeast in the central part of the Appalachian orogen prior to ~ 316 Ma. In the adjacent Blue Ridge–Piedmont thrust sheet, Dallmeyer (1988) interpreted $^{40}Ar/^{39}Ar$ release spectrum ages for hornblende of 330 to 320 Ma to record emplacement of the allochthon and of 320 to 310 Ma to record uplift cooling. Further east in the Kiokee slate belt in the core of the orogen, peak dynamo-thermal metamorphism occurred between 317 and 292 Ma based on dating of deformed and undeformed granites, and between ~ 320 and ~ 295 Ma based on $^{40}Ar/^{39}Ar$ cooling ages of metamorphic minerals (Dallmeyer et al., 1986).

5.6. Timing scenario for MVT mineralisation

Shortly after impact of the African craton with the North American craton in the early Mississippian to initiate the Appalachian orogen (Hatcher, 1989), uplift was already sufficient in its central region by Meramecian time (~ 333 ± 4 Ma) to initiate deposition of the Pennington–Lee clastic wedge in the M-JC district. Uplift was also sufficient to provide a hydrologic head to start to drive fluids out of the adjacent Sevier foreland basin to the east and through the Knox Group in the M-JC district.

Between ~ 330 and ~ 320 Ma, orogenic activity peaked in the M-JC district and is marked by: (1) the greatest depositional rates in the Pennington–Lee clastic wedge (Hatcher et al., 1989); (2) fluid flow giving the 322 Ma $^{40}Ar/^{39}Ar$ date for authigenic K-feldspar (Hearn et al., 1987); and (3) emplacement of the Blue Ridge–Piedmont thrust sheet to within ~ 25 km of the M-JC district (Dallmeyer, 1988), which accommodated as much as 260 km of crustal shortening across the orogen on thrust faults (Cook et al.,1979). As the thrust sheet pushed northwestwards, it would have also pushed against the Valley and

Ridge Province, producing much of the thrust fault transport and related folding in the sheet. During the ~ 330 to ~ 320 Ma interval, it is suggested that the regional dolostone host rocks acquired a chemical remanent magnetisation as their ChRM by fluid flow and/or increase burial pressure causing pre-existing limestone to be dolomitised and/or pre-existing dolostones to be recrystallised. This timing is predicated on the assumption that additional paleomagnetic analyses would lower the 334 ± 14 Ma age for the dolostones ChRM slightly as well as reduce the time span of its error limits.

In this scenario, precipitation of the MVT mineralisation at 316 ± 8 Ma coincides with the waning, rather than initial, stages of the Alleghanian orogeny in M-JC district between ~ 320 and ~ 300 Ma. Its abatement is marked by: (1) termination of sedimentation on the Pennington–Lee wedge by $\sim 307 \pm 3$ Ma; (2) uplift cooling in the adjacent Blue Ridge–Piedmont thrust sheet dated at ~ 320 to ~ 310 Ma by Dallmeyer (1988); and (3) the older of the K–Ar illite dates determined by Sedivy et al. (1984) for the Conasauga shales. The proposed timing for mineralisation is consistent with its paleomagnetic pole, with the late syn-tectonic and post-tectonic fold tests of Bachtadse et al. (1987) and this study, and it is permissible because it does not exceed the ages of the youngest strata cut by the thrust faults in the M-JC region. The ChRM carried by the MVT mineralisation is deemed to be a primary chemical remanent magnetisation acquired mostly upon precipitation during open-space filling.

Also in this scenario, minor post-ore tectonism is permissible in the M-JC district without affecting the paleomagnetic results. Observed minor post-ore vertical displacements may well have occurred on normal and reverse faults as well as residual horizontal motions of several kilometers on sole faults without affecting the paleomagnetic dates significantly. Overall, this scenario is consistent with either the topographically induced fluid flow model for MVT ore genesis favored by Garven et al. (1993) and/or the thrust-induced fluid flow model favored by Oliver (1992). Because the M-JC district is housed within the Appalachian orogen, the latter "squeegee" model cannot be ruled out based on its being too distant for the available driving force on the fluid from compaction.

6. Conclusions

(1) The MVT mineralisation in the M-JC district has a primary chemical remanent magnetisation as its ChRM that is carried by both pyrrhotite and magnetite and that resides in both dolomite and sphalerite. The magnetisation is coeval with MVT mineralisation, post-dates brecciation, and post-dates most or all tilt deformation. The MVT mineralisation is paleomagnetically dated at 316 ± 8 Ma, and is interpreted to have been emplaced after the peak of Alleghanian orogenesis in the M-JC district.

(2) The dolostone host rocks of the M-JC MVT mineralisation have a stable ChRM that is carried by both pyrrhotite and magnetite. It is interpreted to be a remagnetisation that was acquired, most likely, by dolomitisation of limestones or, less likely, by recrystallisation of pre-existing fine-grained dolostones. This magnetisation gives a paleomagnetic date of 334 ± 14 Ma that: (a) requires analyses of additional paleomagnetic specimens to increase its precision; and (b) is thought to have been acquired during peak-Alleghanian orogenesis in the district between ~ 330 and ~ 320 Ma in the initial stages of the same hydrothermal event that led to mineralisation.

(3) The genesis of the MVT mineralisation can be explained by models that invoke either topographically-driven or thrust fault-driven fluid flow in response to Alleghanian orogenesis.

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